

UNIVERSITY INSTITUTE OF ENGINEERING

COMPUTER SCIENCE & ENGINEERING

BACHELOR OF ENGINEERING

CSE/ 4th SEMESTER

OPERATING SYSTEM(24CST-205/24ITT-205)

Unit No.3

Chapter No. 3

Lecture No.42

Topic :System and network threats

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Learning Objectives

- Identify various system and network threats such as viruses, worms, denial-of-service attacks, and unauthorized access.
- Understand how these threats affect system performance, availability, and data security.
- Analyze techniques and mechanisms used to detect, prevent, and mitigate system and network attacks.

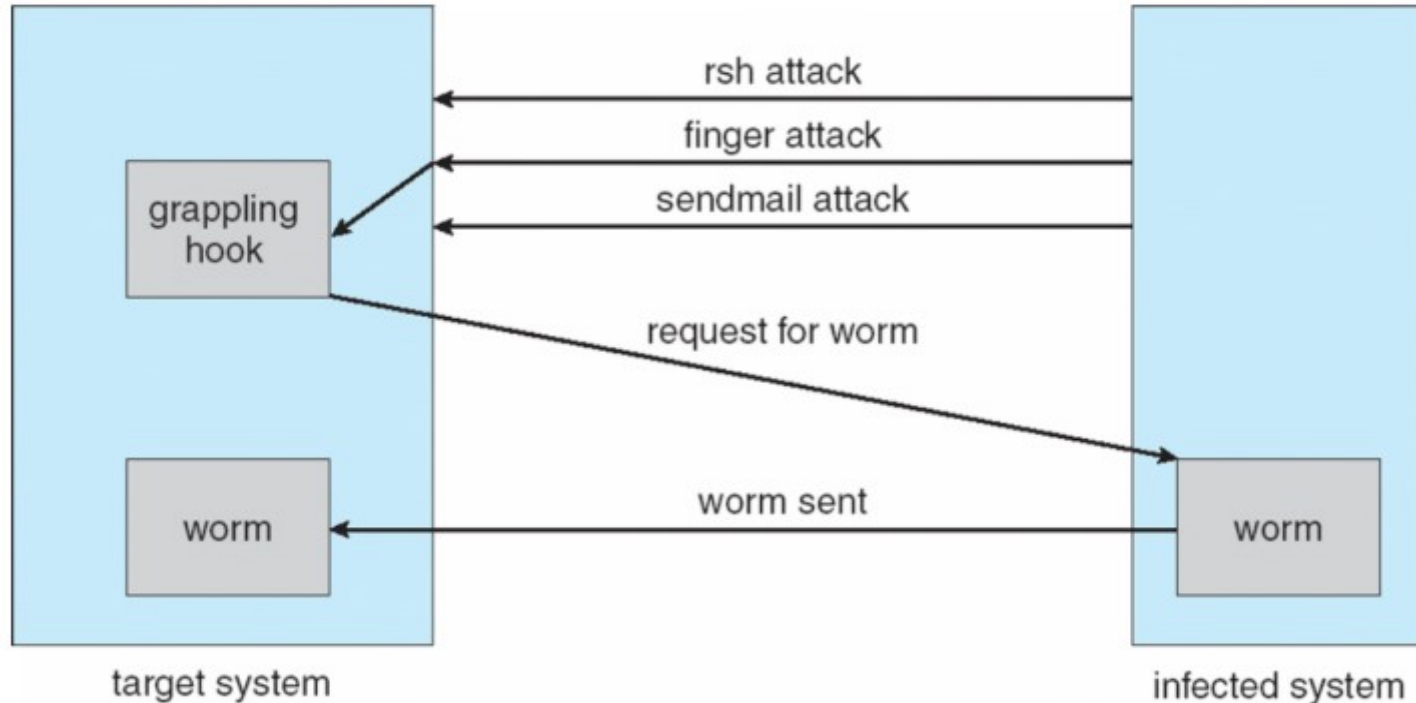
System and Network Threats

- Some systems “open” rather than **secure by default**
 - Reduce attack surface
 - But harder to use, more knowledge needed to administer
- Network threats harder to detect, prevent
 - Protection systems weaker
 - More difficult to have a shared secret on which to base access
 - No physical limits once system attached to internet
 - 4 Or on network with system attached to internet
 - Even determining location of connecting system difficult
 - 4 IP address is only knowledge

System and Network Threats

- **Worms** – use **spawn** mechanism; standalone program
- Internet worm
 - Exploited UNIX networking features (remote access) and bugs in *finger* and *sendmail* programs
 - Exploited trust-relationship mechanism used by *rsh* to access friendly systems without use of password
 - **Grappling hook** program uploaded main worm program
 - 4 99 lines of C code
 - Hooked system then uploaded main code, tried to attack connected systems
 - Also tried to break into other users accounts on local system via password guessing
 - If target system already infected, abort, except for every 7th time

The Morris Internet Worm



System and Network Threats (Cont.)

- **Port scanning**
 - Automated attempt to connect to a range of ports on one or a range of IP addresses
 - Detection of answering service protocol
 - Detection of OS and version running on system
 - `nmap` scans all ports in a given IP range for a response
 - `nessus` has a database of protocols and bugs (and exploits) to apply against a system
 - Frequently launched from **zombie systems**
 - 4 To decrease trace-ability

System and Network Threats (Cont.)

- **Denial of Service**
 - Overload the targeted computer preventing it from doing any useful work
 - **Distributed denial-of-service (DDOS)** come from multiple sites at once
 - Consider the start of the IP-connection handshake (SYN)
 - 4 How many started-connections can the OS handle?
 - Consider traffic to a web site
 - 4 How can you tell the difference between being a target and being really popular?
 - Accidental – CS students writing bad `fork()` code
 - Purposeful – extortion, punishment



Summary of the Lecture

This lecture enables the students to understand the concept of system and network threats, including their types, impact, and methods for detection and prevention.



Next Lecture

Distributed and Network Operating Systems

Quiz/ FAQ's

Q1. Which of the following is a network-based attack?

- a) Trojan horse
- b) Denial-of-Service (DoS) attack
- c) Logic bomb
- d) Trapdoor

Q2. A worm differs from a virus because it:

- a) Requires a host file
- b) Cannot replicate
- c) Spreads independently over networks
- d) Is not harmful

References

<https://www.includehelp.com/c-programming-questions/>

<https://www.studytonight.com/operating-system/>

<https://computing.llnl.gov/tutorials/>

[https://www.tutorialspoint.com/operating_system/index.htm#:~:text=An%20operating%20system%20\(OS\)%20is,software%20in%20a%20computer%20system.](https://www.tutorialspoint.com/operating_system/index.htm#:~:text=An%20operating%20system%20(OS)%20is,software%20in%20a%20computer%20system.)

<https://www.javatpoint.com/os-tutorial>

<https://www.guru99.com/operating-system-tutorial.html>

<https://www.geeksforgeeks.org/operating-systems/>



Faculty-curated videos, NPTEL, Coursera, LinkedIn, or other relevant learning resources

- <https://nptel.ac.in/courses/106105214>
- <https://www.linkedin.com/learning/iot-foundations-operating-systems-fundamentals-2018>



Class-Wise Feedback

Faculty members are encouraged to generate a class specific QR code at the end of each session to collect brief student feedback. This can support self-reflection, improve teaching practices, and allow students to share insights, suggestions, or concerns anonymously.



Thank You

For queries

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